

Battery Powered Photodetector Assembly Instructions

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A battery-powered photodetector design which runs off a 9V battery, with $\pm 12V$ max output. Designed specifically for the Thorlabs EECT3011 enclosure.

Transimpedance Amplifier

This is the main amplification board. Once assembled, it can be tested and verified independently using an external bipolar power supply.

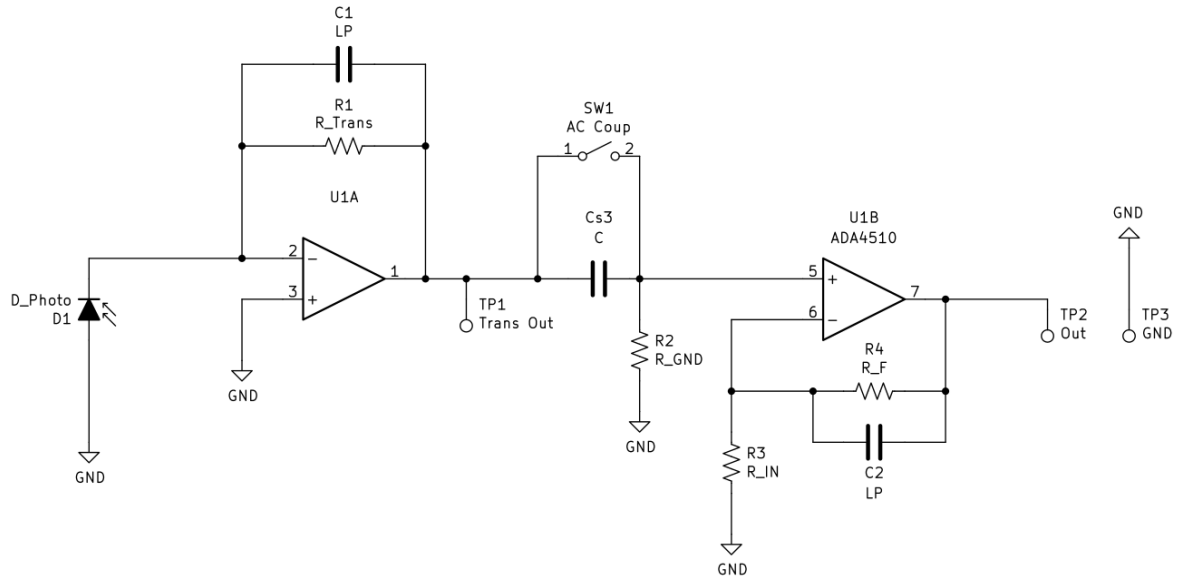


Figure 1: Schematic of transimpedance amplifier board.

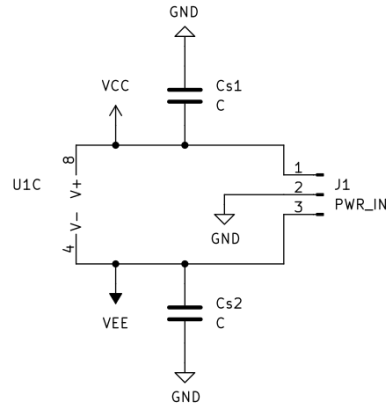


Figure 2: Power Connections and decoupling capacitors. The Cs1 and Cs2 are surface mount capacitors. U1C corresponds to the V_{CC} and V_{EE} connections of the op-amp U1.

Resistor and Capacitor Selection for Transimpedance amplifier

There are three main parameters to choose in this design:

- Transimpedance gain, secondary gain
- Upper Cutoff frequency
- AC Coupling Cutoff frequency (optional)

The selection of these parameters is performed as follows.

Transimpedance Gain

The value of **R_Trans** gives the transimpedance value. For a photocurrent I_p generated by the action of light on the photodiode, the output of the transimpedance amplifier will be

$$V_{\text{Trans}} = I_p \cdot R_{\text{Trans}}$$

There is a second gain stage that can be configured for different gain values. The gain of this second stage is given in terms of **R_F** and **R_IN** as

$$G_2 = 1 + \frac{R_F}{R_{\text{IN}}}$$

This second stage can be optionally configured as a unity gain buffer by omitting **R_IN** and shorting **R_F**. In this case, **C2** can also be omitted.

Note: When AC coupling is used, a gain of $>2\times$ is needed to avoid premature saturation of the first stage when large signals are encountered.

Upper Cutoff Frequency

The upper frequency cutoff is formed by three distinct single-pole filters, formed by the RC time constants of:

- The first stage cutoff frequency, formed by the time constant **R_Trans** $\times$ **C1**
- The second stage cutoff frequency, formed by the time constant **R_F** $\times$ **C2**
- The photodiode internal capacitance, C_p , and the transimpedance: $C_p \times R_{\text{Trans}}$

Each of these three RC time constants form a 1st order filter with cutoff frequency

$$f = \frac{1}{2\pi RC}$$

AC Coupling Frequency

The AC coupling feature of this design is optional and can be bypassed entirely by installing a jumper across **Cs3**. This is also the job of the optional **AC Coup** switch.

When implemented, the AC coupling feature gives a high-pass cutoff frequency of $f = 1/(2\pi \times R_{\text{GND}} \times Cs3)$

A Working Design

The following design is provided in Table 1 as a working starting point. The transimpedance gain can be adjusted for different photodiodes. Sometimes it is possible to eliminate **C1** if the op-amp stability permits. The internal capacitance of the photodiode is not shown here.

Table 1: A generic design for a 220k Ω Transimpedance, $2\times$ gain photodetector. The switchable AC coupling frequency is set to 3.3Hz, and the upper cutoff frequency is roughly 80KHz, but the response remains measurable to at least 150kHz. Tested with a Marktech Optoelectronics MTPD1346D-100 InGaAs photodiode, and a TI-072 op-amp.

R_Trans	C1	Cs3	R_GND	R_IN	R_F	C2
220k Ω	10pF	4.7 μ F	10k Ω	1k Ω	1k Ω	1.5nF

Table 2: Links to Passive Component Sources

Cs3	4.7 μ F Surface Mount Capacitor	Mouser Link
R_IN, R_F	1k Ω Resistor, 1%, 100ppm	Mouser Link
R_GND	10k Ω Resistor, 1%, 50ppm	Mouser Link

Table 3: Operational amplifier IC selection guide for U1.

TI-072	The "Jelly Bean". Cheap, decent noise.	Mouser Link
OPA1678	Lower Noise, Audiophile Grade. Rail-to-Rail Output	Mouser Link

The Amplifier Board

Solder the surface mounts first, using solder paste. After the four surface mount components are soldered, the passives can be hand-soldered with an iron.

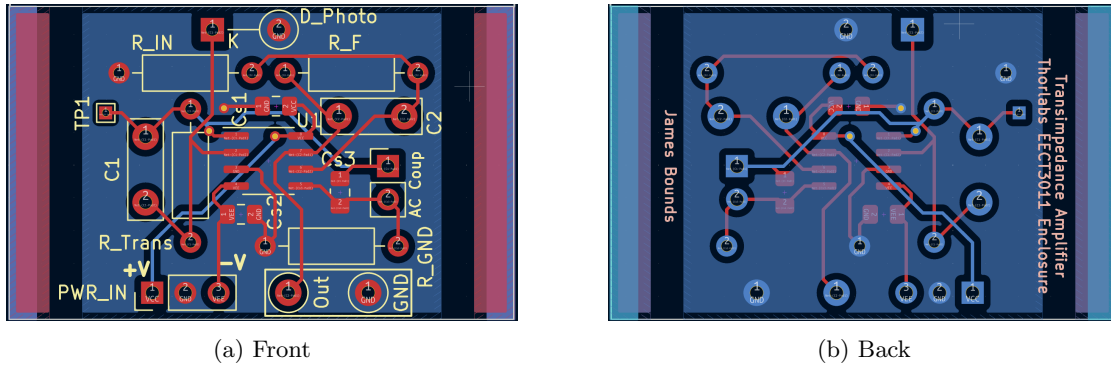


Figure 3: Transimpedance amplifier PCB board.

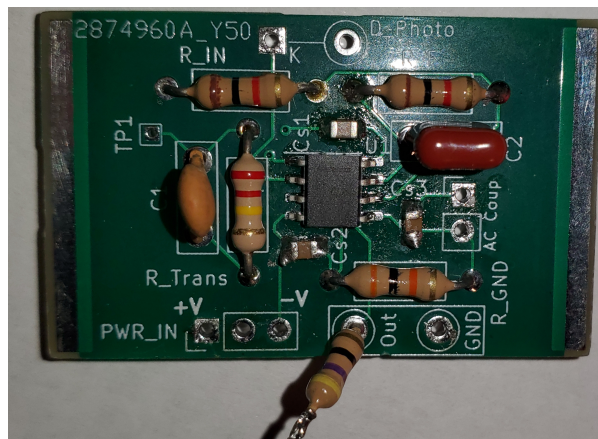


Figure 4: Soldered transimpedance amplifier board. An external 47 Ω resistor is shown connected in series with the output for short-circuit protection and impedance matching with a 50 Ω transmission line.

Testing Notes

The amplifier board can be tested independently from the power supply circuit. This can be done by using an external ± 12 V bipolar power supply.

Power Supply Circuit